Safe Multi-Agent Reinforcement Learning for Robust and Scalable Multi-Vehicle Control in Cooperative Transportation Systems

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***Abstract*—Maintaining safe multi-vehicle coordination in cooperative transportation systems (CTS) is challenging under model errors and variations in vehicle count and payload configuration. In this paper, we propose a novel safe multi-agent reinforcement learning (MARL) framework for scalable multi-vehicle collaborative control in CTS under model uncertainty. Specifically, to improve the modeling accuracy of carrier vehicle dynamics, we develop a disturbance observer that estimates and compensates for model errors online. Based on these estimates, we construct a robust high-order control barrier function safety filter with active coordination to bound coordination errors and suppress their accumulation. Building on this filter, we propose a novel MARL algorithm that integrates it to learn a safe and scalable multi-vehicle collaborative control policy. The learned policy can be reused across CTS with different vehicle counts and payload configurations without retraining. Experiments show that our method reduces coordination error by 76.4% and tracking error by 62.2% compared with the best baseline. It also demonstrates superior robustness to model errors, achieving over 97% task success with coordination errors below 0.01 m across varying model error scales. The same policy achieves 100% task success in scalability tests across unseen CTS configurations. The video and code are available at https://bitmobility.github.io/SSR-MARL/**

***Index Terms*—Cooperative transportation, multi-agent reinforcement learning, robust control barrier functions**

I. Introduction

Cooperative transportation systems (CTS) use multiple carrier vehicles to transport large payloads in construction, manufacturing, and logistics. The shared payload couples vehicle motions, requiring coordinated acceleration and steering to guide the payload along the reference path. As illustrated in Fig. 1(a), inconsistent control actions can disrupt coordination and cause payload damage or vehicle rollover. Model-based control approaches rely on precise system dynamics modeling and handcrafted motion constraints to ensure coordination. However, coupled vehicle–payload dynamics make accurate modeling difficult [1]. Multi-agent reinforcement learning (MARL) learns cooperative policies through interaction with the environment, reducing reliance on detailed system models. However, CTS may employ different numbers and arrangements of carrier vehicles depending on the payload, as shown in Fig. 1(b). Existing MARL methods often target a fixed transportation system, limiting their scalability across transport configurations [2]. Moreover, reward-based policy optimization alone cannot guarantee safe vehicle coordination throughout transportation [3].

Fig. 1. (a) Rollover accidents caused by uncoordinated vehicle motions. (b) Cooperative transportation systems in different configurations (2, 3, and 4 carrier vehicles from left to right).

Safety filters based on control barrier functions (CBFs) incorporate coordination constraints into MARL through online action correction. However, their safety guarantees depend on accurate coupled dynamics modeling and can be undermined by model errors [4]. Moreover, CBF constraints limit coordination errors to an admissible range without actively reducing them. As errors accumulate, the required corrections can exceed the carriers' acceleration or steering limits.

To address these challenges, we propose a novel safe MARL framework for scalable multi-vehicle collaborative control in CTS under model uncertainty. We develop a disturbance observer (DOB) to compensate for the mismatch between nominal model predictions and actual vehicle responses through online error estimation. Subsequently, a robust high-order

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CBF-based safety filter with active coordination is constructed using the DOB estimates to bound coordination errors and actively reduce their accumulation. Finally, we propose a scalable safe MARL algorithm that integrates the safety filter for online action correction and uses the corrections as reward feedback. The algorithm learns a shared control policy across different vehicle counts and transport configurations.

The main contributions are summarized as follows:

- We propose a novel MARL framework for safe, robust, and scalable multi-vehicle collaborative control in CTS. It couples policy learning with a DOB-CBF safety filter, allowing a shared control policy to be applied to CTS with different vehicle counts and payload configurations.
- To reduce the impact of model errors on vehicle coordination, we construct a DOB that estimates acceleration and yaw-rate mismatches online. It corrects the safety filter's motion predictions, allowing the framework to maintain task success rates above 97% and coordination errors below $0.01\,\mathrm{m}$ across different model error scales.
- We propose a robust high-order CBF safety filter with active coordination. It corrects MARL actions to bound coordination errors and actively reduce them within the admissible range. The corrections also guide policy learning through the reward feedback. With this filter, our method achieves 76.4% lower coordination error and 62.2% lower tracking error than the best baseline.

II. Related Work

A. Multi-Vehicle Cooperative Transportation

Cooperative control methods for multi-vehicle transportation are classified as model-based or learning-based. Model-based methods describe coupled vehicle–payload dynamics and geometric constraints to maintain coordinated motion. Constraint-following methods address trajectory tracking and payload-force distribution [5], while adaptive robust control compensates for uncertain dynamics and disturbances [6]. However, reliance on detailed models limits scalability across transport configurations. Learning-based approaches such as MARL learn cooperative policies through rewards and interactions. Decentralized deep Q-network controllers map sensed states to actions, enabling cooperation without predefined leader–follower roles [7]. LDC-MARL combines large language model guidance and coordination constraints through Lagrangian policy optimization to improve learning efficiency and motion coordination [8]. However, safe coordination throughout transportation remains challenging for these learning-based methods.

B. Safe Multi-Agent Reinforcement Learning

Safe MARL addresses safety through constrained policy optimization or action filtering. Lagrangian methods incorporate constraint costs into policy optimization [9], but bounding expected cumulative costs allows individual actions to violate safety constraints. CBF-based methods correct policy actions to satisfy state-dependent safety conditions [10], [11]. However, model errors can cause constraint violations in filters based on nominal dynamics. Robust CBF-based filters address model uncertainty [12], including DOB-based approaches that estimate model errors online and incorporate estimates into CBF constraints [13], [14]. Nevertheless, these methods focus on constraint satisfaction without addressing coordination error reduction within the safe set. Accumulated errors can require corrective inputs beyond control limits. Varying payload geometry and carrier count require scalable policies that maintain coordinated motion across transport configurations [15].

TABLE I
Definitions of the Task Reward Terms

Reward term	**Reward function**
Mission completion	$r_{\mathrm{mission}} = \begin{cases} 20, & \text{task success,} \\ -20, & \text{task failure,} \\ 0, & \text{otherwise} \end{cases}$
Path progress	$r_{\mathrm{progress}} = \ell_{t+1} - \ell_t$
Path tracking	$r_{\mathrm{tracking}} = -0.4\left[(x_p - x_r)^2 + (y_p - y_r)^2\right]$
Coordination	$r_{\mathrm{coordination}} = -\dfrac{0.01}{N}\displaystyle\sum_{i=1}^{N}\left(\dfrac{e_i}{0.1}\right)^2$
Time penalty	$r_{\mathrm{time}} = \begin{cases} -0.1, & t < T-1, \\ 0, & t = T-1 \end{cases}$

III. Problem Formulation

We consider a cooperative transport system in which multiple carrier vehicles transport a common rigid payload along a prescribed path. The vehicles must coordinate their motions to maintain the attachment geometry between the vehicles and the payload throughout transportation. Our goal is to learn a MARL-based control policy that enables safe and coordinated transportation while accurately following the reference path.

The MARL-based control policy is formulated as an N-agent cooperative Markov game, characterized by the tuple:

$$\mathcal{G} = \langle N, \mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma \rangle, \tag{1}$$

where N is the carrier count, $\mathcal{S}$ is the state space, and $\mathcal{A} = \prod_{i=1}^{N} \mathcal{A}_i$ is the joint action space. At step t, each agent i selects $\mathbf{a}_{i,t}$ using the shared policy $\pi_\phi(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})$, where $\mathbf{s}_{i,t}$ is its local state. Given the joint action $\mathbf{a}_t = (\mathbf{a}_{1,t}, \ldots, \mathbf{a}_{N,t})$, the environment transitions to $\mathbf{s}_{t+1}$ according to $\mathcal{P}(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t)$, and all agents receive the shared reward $r_t = r(\mathbf{s}_t, \mathbf{a}_t)$. We maximize the expected cumulative reward:

$$\max_{\phi} J(\phi) = \max_{\phi} \mathbb{E}_{\pi_\phi}\left[\sum_{t=0}^{T-1} \gamma^t r(\mathbf{s}_t, \mathbf{a}_t)\right], \tag{2}$$

where the expectation is taken over trajectories induced by π_ϕ, $\gamma \in (0, 1]$ is the discount factor, and T is the episode length.

A. Formulation of MARL-Based Control Policy

1) State Space: The state space consists of vehicle states, payload states, and waypoint states:

$$\begin{aligned} \mathbf{s}_i^{\mathrm{veh}} &= (v_i, \Delta\theta_i, x_i^m, y_i^m), \\ \mathbf{s}^{\mathrm{pay}} &= (v_x^p, v_y^p, \omega_p, N), \\ \mathbf{s}^{\mathrm{wp}} &= (x_{wp}, y_{wp}, \Delta\theta_{wp}). \end{aligned} \tag{3}$$

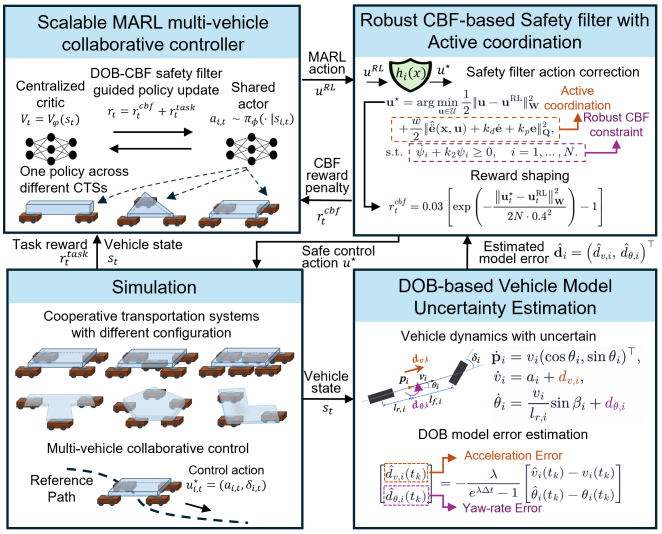

Fig. 2. Overview of the proposed framework.

Here, v_i is carrier i's speed, $\Delta\theta_i$ its heading relative to the payload, and (x_i^m, y_i^m) its fixed attachment position. The payload states include velocity (v_x^p, v_y^p), yaw rate ω_p, and vehicle count N. The waypoint states specify the target position (x_{wp}, y_{wp}) on the reference route and heading $\Delta\theta_{wp}$ relative to the payload. All positions and the payload velocity are expressed in the payload coordinate system.

2) Action Space: The action space consists of acceleration $a_i \in [-1, 1]\,\mathrm{m/s^2}$ and steering angle $\delta_i \in [-\pi/6, \pi/6]\,\mathrm{rad}$.

3) Reward Function: The reward function consists of a task reward r_t^{task} and a safety-correction reward r_t^{cbf}:

$$\begin{aligned} r_t &= r_t^{\mathrm{task}} + r_t^{\mathrm{cbf}}, \\ r_t^{\mathrm{task}} &= r_{\mathrm{mission}} + r_{\mathrm{progress}} + r_{\mathrm{tracking}} + r_{\mathrm{coordination}} + r_{\mathrm{time}}. \end{aligned} \tag{4}$$

The five task reward terms are defined in Table I, where ℓ_t is the payload's projected arc length along the reference path. (x_p, y_p) and (x_r, y_r) denote the payload center and closest reference waypoint. The coordination error e_i measures deviation from the attachment position transformed into global coordinates. The safety-correction reward penalizes filter interventions (Section IV-C).

IV. Proposed Framework

As shown in Fig. 2, a robust high-order CBF safety filter uses DOB estimates to correct actions and bound coordination errors. Active coordination suppresses error accumulation. The corrections provide reward feedback for learning a shared policy across vehicle counts and payload configurations.

A. DOB-Based Vehicle Model Uncertainty Estimation

Each carrier is described by a simplified kinematic bicycle model. Let $\mathbf{p}_i = (x_i, y_i)^\top$ and θ_i denote its position and heading in the global coordinate system. Unmodeled dynamics introduce mismatch between the nominal and actual vehicle responses. This mismatch causes errors in predicted vehicle speeds and headings, which can impair carrier coordination. We represent these model errors by additive uncertainty terms and write the vehicle dynamics as

$$\begin{aligned} \dot{\mathbf{p}}_i &= v_i(\cos\theta_i, \sin\theta_i)^\top, \\ \dot{v}_i &= a_i + d_{v,i}, \\ \dot{\theta}_i &= \frac{v_i}{l_{r,i}} \sin\beta_i + d_{\theta,i}, \end{aligned} \tag{5}$$

where $\beta_i = \arctan\!\left(\frac{l_{r,i}}{l_{f,i}+l_{r,i}} \tan\delta_i\right)$, and $l_{f,i}$ and $l_{r,i}$ denote distances from the vehicle center to the front and rear axles, respectively. $d_{v,i}$ and $d_{\theta,i}$ capture differences between nominal and actual longitudinal acceleration and yaw rate.

To compensate for these prediction errors, we construct a DOB that estimates unmodeled acceleration and yaw-rate terms online and corrects nominal predictions. The DOB comprises a state predictor and a piecewise-constant estimation law. For vehicle i, the predictor uses applied acceleration and steering angle to predict its speed and heading:

$$\begin{aligned} \dot{\hat{v}}_i &= a_i + \hat{d}_{v,i} - \lambda(\hat{v}_i - v_i), \\ \dot{\hat{\theta}}_i &= \frac{v_i}{l_{r,i}} \sin\beta_i + \hat{d}_{\theta,i} - \lambda(\hat{\theta}_i - \theta_i), \end{aligned} \tag{6}$$

where $\hat{v}_i$ and $\hat{\theta}_i$ are predicted speed and heading, $\hat{d}_{v,i}$ and $\hat{d}_{\theta,i}$ are estimated acceleration and yaw-rate errors, and $\lambda > 0$ is the observer gain. The estimates correct the nominal vehicle response. Initially, the predicted speed and heading equal their measured values, and both uncertainty estimates are zero.

The two uncertainty estimates are collected as $\hat{\mathbf{d}}_i = (\hat{d}_{v,i}, \hat{d}_{\theta,i})^\top$. At each sampling instant $t_k = k\Delta t$, they are updated using speed and heading prediction errors and held constant between updates:

$$\begin{aligned} \hat{\mathbf{d}}_i(t_k) &= -\frac{\lambda}{e^{\lambda\Delta t} - 1} \begin{bmatrix} \hat{v}_i(t_k) - v_i(t_k) \\ \hat{\theta}_i(t_k) - \theta_i(t_k) \end{bmatrix}, \\ \hat{\mathbf{d}}_i(t) &= \hat{\mathbf{d}}_i(t_k), \quad t \in [t_k, t_{k+1}), \end{aligned} \tag{7}$$

Between updates, the DOB estimates remain constant while the actual uncertainty can vary with the vehicle motion and control inputs. To account for the resulting estimation errors in the coordination constraints, we assume the following bounds throughout each control interval:

$$|d_{v,i}(t) - \hat{d}_{v,i}(t)| \le \varepsilon_v, \qquad |d_{\theta,i}(t) - \hat{d}_{\theta,i}(t)| \le \varepsilon_\theta. \tag{8}$$

The bounds ε_v and ε_θ cover the residual errors associated with initialization, estimation delay, and numerical integration. The estimates and these bounds are incorporated into the coordination constraints in Section IV-B.

B. Robust CBF-Based Safety Filter with Active Coordination

For vehicle i, let $\mathbf{x}_i = (x_i, y_i, v_i, \theta_i)^\top$ and $\mathbf{u}_i = (a_i, \sin\beta_i)^\top$. The dynamics in (5) can be expressed in control-affine form as

$$\dot{\mathbf{x}}_i = \underbrace{\begin{bmatrix} v_i\cos\theta_i \\ v_i\sin\theta_i \\ 0 \\ 0 \end{bmatrix}}_{f_i(\mathbf{x}_i)} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & v_i/l_{r,i} \end{bmatrix}}_{g_i(\mathbf{x}_i)} \mathbf{u}_i + \begin{bmatrix} 0 \\ 0 \\ d_{v,i} \\ d_{\theta,i} \end{bmatrix}, \tag{9}$$

where f_i describes the nominal vehicle motion and g_i maps the acceleration and transformed steering input to the speed and heading dynamics. Stacking all vehicle states and inputs gives the joint system $\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} + \mathbf{d}$. The disturbance $\mathbf{d}$, its DOB estimate $\hat{\mathbf{d}}$, and the error bound $\boldsymbol{\varepsilon}$ are arranged consistently with the joint state.

1) Robust High-Order CBF: The coordination error and barrier function for carrier i are defined as

$$\begin{aligned} \mathbf{e}_i(\mathbf{x}) &= \mathbf{p}_i - \mathbf{p}_p - R(\theta_p)\mathbf{r}_i, \\ h_i(\mathbf{x}) &= e_{\max}^2 - \|\mathbf{e}_i(\mathbf{x})\|^2, \end{aligned} \tag{10}$$

where $\mathbf{p}_p = (x_p, y_p)^\top$ and θ_p are the payload position and heading, $\mathbf{r}_i = (x_i^m, y_i^m)^\top$ locates carrier i's attachment point in the payload frame. $R(\theta_p)$ rotates this vector into the global frame. $e_{\max}$ is the admissible error limit. Positive, zero, and negative values of h_i indicate coordination errors below, at, and above the admissible limit, respectively. The safe set is

$$\mathcal{C} = \{\mathbf{x} : h_i(\mathbf{x}) \geq 0,\ i = 1, \ldots, N\}. \tag{11}$$

To maintain coordination during motion, we impose the second-order CBF conditions:

$$\begin{aligned} \psi_i &= \dot{h}_i + k_1 h_i \geq 0, \\ \dot{\psi}_i &+ k_2 \psi_i \geq 0, \end{aligned} \tag{12}$$

where $k_1, k_2 > 0$. The first inequality limits the decrease of the safety margin. Given $h_i(0) \geq 0$ and $\psi_i(0) \geq 0$, the second preserves $\psi_i \geq 0$ and $h_i \geq 0$.

Model uncertainty affects the barrier function's second derivative. Incorporating the DOB estimate and error bounds into the second inequality of (12) yields

$$\begin{aligned} &L_f^2 h_i + L_g L_f h_i\, \mathbf{u} + \nabla(L_f h_i)^\top \hat{\mathbf{d}} \\ &\quad - \left|\nabla(L_f h_i)\right|^\top \boldsymbol{\varepsilon} + (k_1 + k_2) L_f h_i + k_1 k_2 h_i \geq 0, \end{aligned} \tag{13}$$

Equation (13) defines the DOB-based robust high-order CBF constraint. It is affine in $\mathbf{u}$ and imposed jointly for all carriers. The robust CBF constraint is derived in the Appendix, together with the conditions for forward invariance.

2) Safety Filter with Active Coordination: CBF constraints bound coordination errors but do not actively reduce them. Accumulating errors may require corrections beyond acceleration or steering limits, making the filter infeasible. We therefore introduce active coordination to reduce errors and damp their dynamics within the safe set.

At each control step, the filter combines action correction and active coordination by solving

$$\begin{aligned} \mathbf{u}^\star = \arg\min_{\mathbf{u}} \quad & \frac{1}{2}\|\mathbf{u} - \mathbf{u}^{\mathrm{RL}}\|_{\mathbf{W}}^2 \\ & + \frac{w}{2}\|\hat{\ddot{\mathbf{e}}}(\mathbf{x}, \mathbf{u}) + k_d \dot{\mathbf{e}} + k_p \mathbf{e}\|_{\mathbf{Q}}^2, \\ \text{s.t.} \quad & (13), \quad i = 1, \ldots, N, \\ & \mathbf{u} \in \mathcal{U}, \end{aligned} \tag{14}$$

where $\mathbf{u}^{\mathrm{RL}}$ is the RL action, $\mathbf{e} \in \mathbb{R}^{2N}$ collects all carriers' coordination errors, and $\hat{\ddot{\mathbf{e}}}(\mathbf{x}, \mathbf{u})$ is the error acceleration predicted by the DOB-compensated model.

The first term penalizes action changes; the second encourages error reduction even when CBF constraints are inactive. The gains $k_p, k_d > 0$ control correction strength and damping, respectively. The matrices $\mathbf{W}$ and $\mathbf{Q}$ are positive diagonal weights, and $w > 0$ weights active coordination. $\mathcal{U}$ enforces acceleration, steering, and speed limits. The problem is a quadratic program because the predicted error acceleration is affine in $\mathbf{u}$. The Appendix details the error acceleration prediction for active coordination and the input bounds.

C. Scalable Safe MARL with DOB-CBF Safety Filter

1) Scalable Actor–Critic Architecture: We use a shared actor and centralized critic. Each carrier applies $\pi_\phi(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})$ to its vehicle, payload, and waypoint states in Section III-A, keeping the actor's input dimension and parameters fixed across configurations. The critic uses a shared encoder to map vehicle states to feature vectors, then computes their element-wise mean and maximum across carriers. These vectors are concatenated with payload and waypoint states to estimate the expected team return V_t from a fixed-dimensional input, accommodating different team sizes.

2) Safe MARL Algorithm with a Safety Filter: At each control step, the DOB-CBF safety filter corrects $\mathbf{u}_t^{\mathrm{RL}}$ through (14). We penalize these corrections to encourage RL actions requiring less intervention:

$$\begin{aligned} c_t &= \frac{1}{2N}\left\|\mathbf{u}_t^\star - \mathbf{u}_t^{\mathrm{RL}}\right\|_{\mathbf{W}}^2, \\ r_t^{\mathrm{cbf}} &= 0.03\left[\exp\!\Big(-\frac{c_t}{0.4^2}\Big) - 1\right], \end{aligned} \tag{15}$$

where c_t is the mean squared correction normalized by input limits through $\mathbf{W}$. Averaging over $2N$ components keeps penalties comparable across team sizes. The penalty is zero for unchanged actions and enters the shared reward in (4).

The actor and critic use trajectories collected with the safety filter. The actor maximizes the entropy-regularized proximal policy optimization (PPO) objective:

$$\begin{aligned} J_{\mathrm{actor}}(\phi) &= \mathbb{E}_t\left[\frac{1}{N}\sum_{i=1}^{N}\Big(\min\{\rho_{i,t}\hat{A}_t, \bar{\rho}_{i,t}\hat{A}_t\} + \alpha H_{i,t}\Big)\right], \\ \bar{\rho}_{i,t} &= \mathrm{clip}(\rho_{i,t}, 1 - \epsilon_\pi, 1 + \epsilon_\pi), \end{aligned} \tag{16}$$

where $\rho_{i,t} = \pi_\phi(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t}) / \pi_{\phi_{\mathrm{old}}}(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})$ is the probability ratio for the sampled RL action, $\hat{A}_t$ is the normalized team advantage, and ϵ_π controls clipping. Policy entropy $H_{i,t} = \mathcal{H}(\pi_\phi(\cdot \mid \mathbf{s}_{i,t}))$ encourages exploration with weight $\alpha > 0$. Averaging over carriers keeps the objective comparable across team sizes. The critic minimizes the squared error against the return target:

$$L_{\mathrm{critic}} = \mathbb{E}_t\left[\Big(V_t - \hat{R}_t\Big)^2\right], \tag{17}$$

where $\hat{R}_t$ is computed from shared rewards and value estimates using generalized advantage estimation.

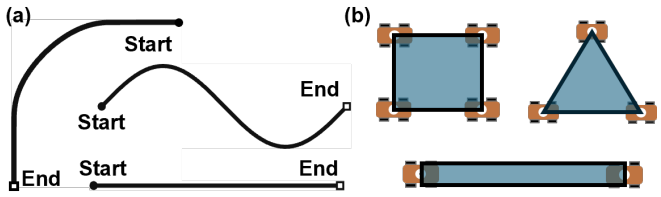

Fig. 3. Reference paths and transport configurations: (a) straight, left-turn, and sinusoidal paths; (b) square, triangular, and linear configurations with four, three, and two carriers, respectively.

V. Experimental Settings

A. Experimental Setup

We evaluate our approach in a custom Gymnasium environment for cooperative payload transportation. As shown in Fig. 3, two, three, and four carriers transport a payload in linear, triangular, and square configurations, respectively, along straight, left-turn, or sinusoidal reference paths. The nine configuration–path combinations are sampled equally during training. All carriers start at $2\,\mathrm{m/s}$ with a maximum speed of $4\,\mathrm{m/s}$. The coordination error limit is $e_{\max} = 0.1\,\mathrm{m}$. Control intervals are $0.1\,\mathrm{s}$ and each episode lasts at most $120\,\mathrm{s}$. We model uncertainty by independently sampling each carrier's acceleration gain, yaw-rate gain, and linear drag coefficient uniformly from $[0.70, 1.30]$, $[0.80, 1.20]$, and $[0.02, 0.08]\,\mathrm{s}^{-1}$, respectively. These parameters remain fixed per episode.

B. Baselines and Implementation Details

We compare our method with three MARL baselines:

1) Independent PPO (IPPO) [16]: IPPO updates each agent using local observations and a local value function.

2) Multi-Agent Deep Deterministic Policy Gradient (MADDPG) [17]: MADDPG optimizes deterministic actors using centralized critics that evaluate joint states and actions.

3) MAPPO-Lagrangian (MAPPO-Lag) [18]: MAPPO-Lag incorporates coordination constraint costs into the clipped MAPPO objective through an adaptive Lagrange multiplier.

All methods share actor parameters across carriers and use identical local actor inputs. Training uses five random seeds, eight parallel environments, and 1.2×10^6 environment steps per seed. All networks use Adam with a learning rate of 3×10^{-4}. Our method collects 128 steps per environment per update. The discount factor, GAE parameter, PPO clipping parameter, and entropy coefficient are 0.995, 0.95, 0.2, and 0.01, respectively. The DOB uses $\lambda = 1\,\mathrm{s}^{-1}$ with estimation-error bounds $\varepsilon_v = 0.7\,\mathrm{m/s}^2$ and $\varepsilon_\theta = 1.8\,\mathrm{rad/s}$. We set $k_1 = k_2 = 4\,\mathrm{s}^{-1}$ for the CBF and $k_p = 1\,\mathrm{s}^{-2}$, $k_d = 2\,\mathrm{s}^{-1}$, and $w = 30$ for active coordination. Experiments are conducted on a platform equipped with an NVIDIA Quadro RTX 8000 GPU.

C. Evaluation Metrics

We evaluate all methods using the following metrics:

1) Episode Return (ER): ER is the undiscounted task reward accumulated over an episode.

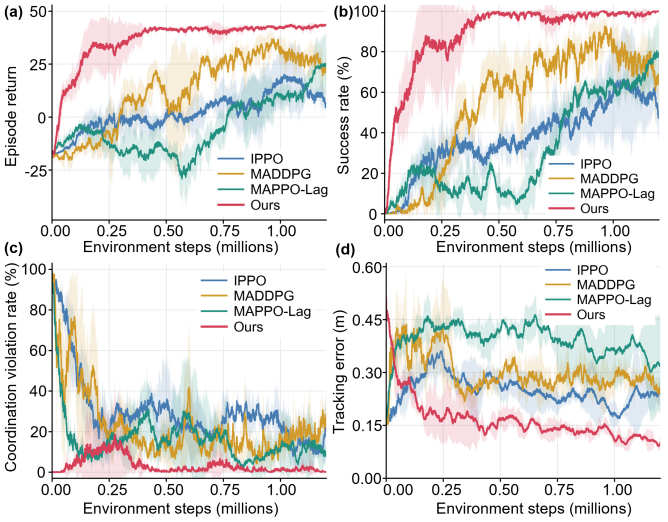

Fig. 4. Training performance comparison: (a) episode return; (b) success rate; (c) coordination violation rate; and (d) tracking errors. The mean and standard deviation for each episode are represented by lines and shadows, respectively.

TABLE II
Test Performance Comparison.

Method	Return ↑	SR (%) ↑	CE (cm) ↓	TE (cm) ↓
IPPO	$14.48_{\pm 4.38}$	$61.85_{\pm 5.25}$	$1.57_{\pm 0.23}$	$22.92_{\pm 2.62}$
MADDPG	$24.68_{\pm 5.34}$	$70.37_{\pm 12.88}$	$3.88_{\pm 0.84}$	$28.29_{\pm 0.49}$
MAPPO-Lag	$29.55_{\pm 26.85}$	$76.30_{\pm 29.13}$	$2.03_{\pm 0.59}$	$31.35_{\pm 14.78}$
Ours	$\mathbf{43.74_{\pm 0.13}}$	$\mathbf{100.00_{\pm 0.00}}$	$\mathbf{0.37_{\pm 0.05}}$	$\mathbf{8.66_{\pm 2.55}}$

2) Success Rate (SR): SR is the percentage of episodes that complete transportation within the time limit without path deviation or coordination constraint violations.

3) Coordination Violation Rate (CVR): CVR is the percentage of episodes in which at least one carrier's coordination error exceeds $e_{\max}$.

4) Coordination Error (CE): CE is the time-averaged root mean square coordination error across carriers, $\mathrm{CE} = \frac{1}{T}\sum_{t=1}^{T}\sqrt{\frac{1}{N}\sum_{i=1}^{N} e_{i,t}^2}$, where $e_{i,t}$ is the coordination error of carrier i at step t.

5) Tracking Error (TE): TE measures the root mean square path-tracking error over an episode, $\mathrm{TE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} d_t^2}$, where d_t is the shortest distance from the payload center to the reference path at step t.

VI. Results and Discussion

We compare training and test performance against MARL baselines, then evaluate robustness to model uncertainty and scalability across vehicle counts and transport configurations. Finally, ablation studies assess the contributions of the CBF safety filter, DOB compensation, and active coordination.

A. Training and Testing Performance Comparison

Fig. 4 shows faster ER and SR improvement for our method than for the three baselines. SR approaches 100% after approximately 0.5×10^6 environment steps. In later training,

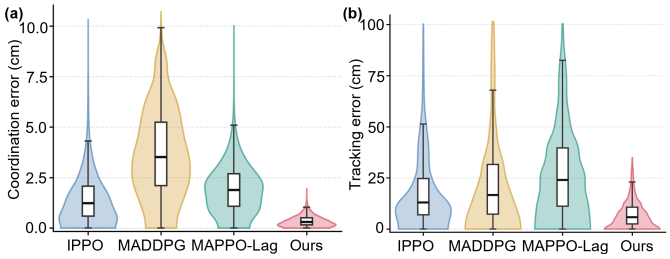

Fig. 5. Distributions of per-step test errors: (a) coordination error of individual carriers; (b) payload path-tracking error. Boxes and central lines indicate the interquartile ranges and medians, respectively.

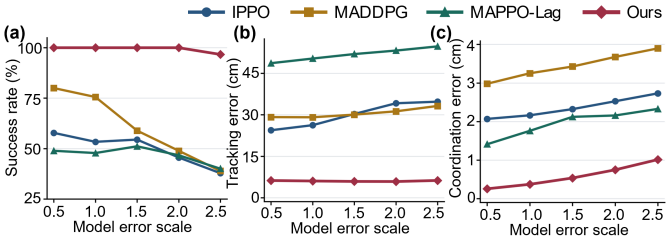

Fig. 6. Testing performance at different model error scales: (a) success rate; (b) tracking error; and (c) coordination error. A model error scale of 1.0 corresponds to the training parameter distributions.

CVR approaches zero and TE decreases to approximately $0.1\,\mathrm{m}$. Table II shows that our method achieves the highest return and SR, exceeding MAPPO-Lag by 23.70 percentage points in SR. Its CE of $0.37\,\mathrm{cm}$ and TE of $8.66\,\mathrm{cm}$ are 76.4% and 62.2% lower than those of IPPO. Fig. 5 presents both error distributions across all testing time steps, with lower medians and greater concentration near zero for our method. These improvements can be attributed to DOB-compensated action correction, active coordination within the safe set, and reward feedback that penalizes large corrections.

B. Robustness to Model Uncertainty

We evaluate the trained policies on the nine configuration–path combinations at model error scales of 0.5, 1.0, 1.5, 2.0, and 2.5. The scale multiplies deviations of the acceleration gain, yaw-rate gain, and linear drag coefficient from their nominal values. Fig. 6 shows that increasing the model error scale generally reduces SR and increases TE and CE. Our method maintains 100% SR up to a scale of 2.0 and approximately 97% at 2.5, where baseline SRs fall to around 40%. Both TE and CE of our method are lower than those of the baselines across all tested scales.

This robustness can be attributed to DOB compensation and robust CBF constraints. The DOB corrects acceleration and yaw-rate predictions online, while the robust CBF tightens constraints using estimation-error bounds to reduce the impact of residual errors on vehicle coordination. These results demonstrate reliable transportation and accurate motion coordination at larger model error scales.

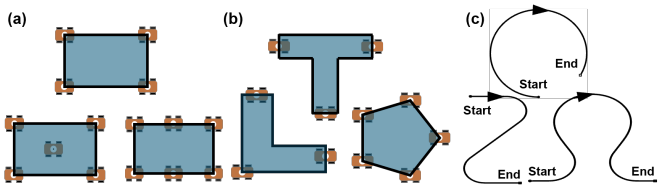

Fig. 7. Scalability test scenarios: (a) rectangular configurations with four to six carriers; (b) T-shaped, L-shaped, and pentagonal configurations; (c) circular-arc, Z-shaped, and Ω-shaped reference paths.

C. Scalability Evaluation

We compare our method and baselines with four, five, and six carriers transporting the same rectangular payload (Fig. 7(a)). We then assess scalability across pentagonal, T-shaped, and L-shaped payloads (Fig. 7(b)). Both tests use unseen circular-arc, Ω-shaped, and Z-shaped paths in Fig. 7(c), without retraining.

Table III shows that our method maintained 100% SR and zero CVR with four to six carriers, while the best baseline reached 10.0% SR with six carriers. Across all three payload shapes, our method again achieved 100% SR and zero CVR, with the lowest CE and TE (Table IV). This performance can be explained by attachment positions in policy inputs and CBF constraints. These positions encode the geometry governing relative motion during turning, allowing the filter to accommodate configuration changes using DOB-compensated predictions. Table V shows that for the rectangular payload, total runtime increased from 1.591 to $1.715\,\mathrm{ms}$ as the team grew from four to six carriers, mainly because the safety filter solves a larger optimization problem. The T-shaped and L-shaped cases took about $1.62\,\mathrm{ms}$, and the pentagonal case took $1.685\,\mathrm{ms}$, owing to its larger optimization problem. Across configurations, MARL computation remains near $0.25\,\mathrm{ms}$ and total time below $1.72\,\mathrm{ms}$, within the $100\,\mathrm{ms}$ control interval.

These results demonstrate that our method accommodates larger teams and different payload shapes without retraining, maintaining successful and coordinated transportation on unseen paths at low computational cost.

D. Ablation Studies

To assess component contributions, we compare the full method with three variants: 1) *w/o CBF Safety Filter* bypasses the safety filter; 2) *w/o DOB Compensation* uses nominal CBF constraints without DOB compensation; and 3) *w/o Active Coordination* removes the active coordination objective while retaining robust CBF constraints.

Fig. 8 shows faster learning and the highest final ER and SR for the full method, with CVR approaching zero. Removing the CBF filter yields the lowest final SR and highest TE; omitting DOB produces the highest CVR in later training. *w/o Active Coordination* ranks second in final ER and SR. In Table VI, removing the filter reduces SR from 100% to 75.93%, reflecting the loss of action correction that supports coordination learning. Without DOB, CVR rises from 0% to 21.85%, as acceleration and yaw-rate prediction errors impair

TABLE III
TEST PERFORMANCE FOR A RECTANGULAR PAYLOAD WITH DIFFERENT VEHICLE COUNTS.

Method	Rectangle 4 vehicles SR (%) ↑	CVR (%) ↓	CE (cm) ↓	TE (cm) ↓	Rectangle 5 vehicles SR (%) ↑	CVR (%) ↓	CE (cm) ↓	TE (cm) ↓	Rectangle 6 vehicles SR (%) ↑	CVR (%) ↓	CE (cm) ↓	TE (cm) ↓
IPPO	$5.6_{\pm 3.8}$	$58.9_{\pm 13.9}$	$2.83_{\pm 0.22}$	$26.99_{\pm 4.72}$	$1.1_{\pm 1.9}$	$71.1_{\pm 13.9}$	$3.00_{\pm 0.22}$	$30.26_{\pm 3.40}$	$0.0_{\pm 0.0}$	$64.4_{\pm 15.0}$	$3.08_{\pm 0.13}$	$30.75_{\pm 3.92}$
MADDPG	$16.7_{\pm 12.0}$	$50.0_{\pm 25.2}$	$4.13_{\pm 0.44}$	$32.92_{\pm 9.14}$	$11.1_{\pm 10.7}$	$60.0_{\pm 29.6}$	$4.00_{\pm 0.50}$	$30.49_{\pm 6.78}$	$10.0_{\pm 14.5}$	$73.3_{\pm 23.3}$	$3.91_{\pm 0.32}$	$26.28_{\pm 11.28}$
MAPPO-Lag	$16.7_{\pm 15.3}$	$57.8_{\pm 30.2}$	$3.49_{\pm 0.55}$	$35.84_{\pm 20.60}$	$6.7_{\pm 5.8}$	$67.8_{\pm 38.5}$	$3.44_{\pm 0.38}$	$44.89_{\pm 17.24}$	$4.4_{\pm 5.1}$	$67.8_{\pm 41.4}$	$3.26_{\pm 0.35}$	$51.88_{\pm 12.46}$
Ours	$\mathbf{100.0_{\pm 0.0}}$	$\mathbf{0.0_{\pm 0.0}}$	$\mathbf{0.55_{\pm 0.08}}$	$\mathbf{11.46_{\pm 4.13}}$	$\mathbf{100.0_{\pm 0.0}}$	$\mathbf{0.0_{\pm 0.0}}$	$\mathbf{0.57_{\pm 0.06}}$	$\mathbf{13.98_{\pm 7.74}}$	$\mathbf{100.0_{\pm 0.0}}$	$\mathbf{0.0_{\pm 0.0}}$	$\mathbf{0.60_{\pm 0.04}}$	$\mathbf{17.53_{\pm 11.28}}$

TABLE IV
TEST PERFORMANCE FOR DIFFERENT PAYLOAD SHAPES.

Method	Pentagon 5 vehicles SR (%) ↑	CVR (%) ↓	CE (cm) ↓	TE (cm) ↓	T-shape 3 vehicles SR (%) ↑	CVR (%) ↓	CE (cm) ↓	TE (cm) ↓	L-shape 3 vehicles SR (%) ↑	CVR (%) ↓	CE (cm) ↓	TE (cm) ↓
IPPO	$0.0_{\pm 0.0}$	$72.2_{\pm 20.4}$	$3.22_{\pm 0.22}$	$29.52_{\pm 3.61}$	$14.4_{\pm 13.5}$	$54.4_{\pm 28.3}$	$2.89_{\pm 0.29}$	$25.69_{\pm 6.79}$	$2.2_{\pm 1.9}$	$50.0_{\pm 12.0}$	$2.81_{\pm 0.10}$	$27.61_{\pm 5.63}$
MADDPG	$18.9_{\pm 17.1}$	$44.4_{\pm 37.5}$	$3.71_{\pm 0.34}$	$32.82_{\pm 3.52}$	$21.1_{\pm 31.0}$	$31.1_{\pm 27.1}$	$3.64_{\pm 0.64}$	$34.29_{\pm 8.78}$	$4.4_{\pm 7.7}$	$21.1_{\pm 19.2}$	$3.88_{\pm 0.36}$	$34.74_{\pm 6.37}$
MAPPO-Lag	$13.3_{\pm 20.3}$	$55.6_{\pm 22.7}$	$3.25_{\pm 0.34}$	$47.73_{\pm 12.67}$	$43.3_{\pm 43.3}$	$25.6_{\pm 18.4}$	$3.09_{\pm 0.19}$	$33.19_{\pm 20.64}$	$22.2_{\pm 30.2}$	$32.2_{\pm 20.1}$	$3.11_{\pm 0.10}$	$34.40_{\pm 17.60}$
Ours	$\mathbf{100.0_{\pm 0.0}}$	$\mathbf{0.0_{\pm 0.0}}$	$\mathbf{0.56_{\pm 0.01}}$	$\mathbf{14.81_{\pm 8.15}}$	$\mathbf{100.0_{\pm 0.0}}$	$\mathbf{0.0_{\pm 0.0}}$	$\mathbf{0.51_{\pm 0.07}}$	$\mathbf{10.48_{\pm 6.27}}$	$\mathbf{100.0_{\pm 0.0}}$	$\mathbf{0.0_{\pm 0.0}}$	$\mathbf{0.49_{\pm 0.07}}$	$\mathbf{11.37_{\pm 1.16}}$

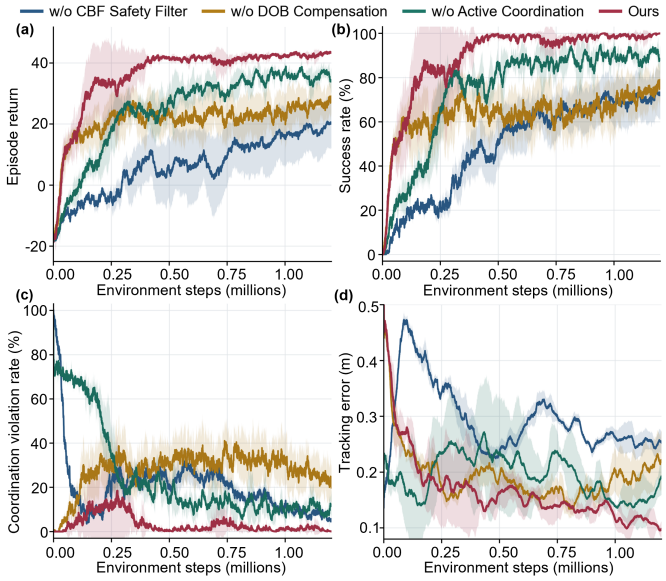

Fig. 8. Training performance of the ablation variants: (a) episode return; (b) success rate; (c) coordination violation rate; and (d) tracking error.

TABLE V
COMPUTATION TIME PER CONTROL STEP FOR OUR METHOD.

Payload shape	Vehicle count	Total (ms)	Safety filter (ms)	MARL (ms)
Rectangle	4	1.591	1.338	0.246
Rectangle	5	1.652	1.398	0.246
Rectangle	6	1.715	1.459	0.248
Pentagon	5	1.685	1.430	0.248
T-shaped	3	1.616	1.361	0.248
L-shaped	3	1.614	1.359	0.248

TABLE VI
ABLATION TEST PERFORMANCE.

Variant	SR (%) ↑	CVR (%) ↓	CE (cm) ↓	TE (cm) ↓
w/o CBF Safety Filter	$75.93_{\pm 36.92}$	$3.70_{\pm 2.31}$	$2.70_{\pm 0.67}$	$23.45_{\pm 7.32}$
w/o DOB Compensation	$76.67_{\pm 19.47}$	$21.85_{\pm 18.34}$	$2.98_{\pm 0.50}$	$23.37_{\pm 12.76}$
w/o Active Coordination	$90.00_{\pm 6.19}$	$10.00_{\pm 6.19}$	$1.80_{\pm 0.22}$	$20.10_{\pm 3.95}$
Ours	$\mathbf{100.00_{\pm 0.00}}$	$\mathbf{0.00_{\pm 0.00}}$	$\mathbf{0.37_{\pm 0.05}}$	$\mathbf{8.66_{\pm 2.55}}$

the filter's assessment of action effects. Removing active coordination increases CE from 0.37 to 1.80 cm and TE from 8.66 to 20.10 cm. CBF constraints permit nonzero deviations that can accumulate until required corrections exceed control limits, making the filter infeasible. Together, the three components improve learning, task completion, and coordination.

VII. CONCLUSION

We present a scalable, robust and safe MARL framework for cooperative payload transportation under model uncertainty. This framework pairs a shared actor to accommodate varying CTS configurations. It incorporates a DOB-compensated CBF safety filter that bounds coordination errors, suppresses their accumulation through active coordination, and supplies action corrections for reward shaping. In simulation, the framework reduced coordination and tracking errors by 76.4% and 62.2%, respectively, relative to IPPO, the best baseline on both metrics. Task success remained approximately 97% at the largest model error scale. Without retraining, the shared policy achieved 100% success across scalability tests with unseen paths, varied payload configurations, and up to six carriers. Future work will validate the framework on a physical transportation platform.

APPENDIX

A. Barrier Derivatives and Robust Constraint

Using the state ordering in (9), the joint disturbance is $\mathbf{d} = (0, 0, d_{v,1}, d_{\theta,1}, \ldots, 0, 0, d_{v,N}, d_{\theta,N})^\top$. Its estimate $\hat{\mathbf{d}}$ uses the corresponding DOB outputs in the same order, and $\boldsymbol{\varepsilon} = (0, 0, \varepsilon_v, \varepsilon_\theta, \ldots, 0, 0, \varepsilon_v, \varepsilon_\theta)^\top$. The payload pose is obtained by minimizing $\sum_i \|\mathbf{p}_i - \mathbf{p}_p - R(\theta_p)\mathbf{r}_i\|^2$ over $\mathbf{p}_p$ and θ_p. Derivatives of $\mathbf{e}_i$ and h_i include this dependence on all vehicle positions. Since the input and disturbance channels act on speed and heading, $L_g h_i = 0$ and $\nabla h_i^\top \mathbf{d} = 0$. Consequently,

$$\dot{h}_i = L_f h_i, \qquad \ddot{h}_i = L_f^2 h_i + L_g L_f h_i \, \mathbf{u} + \nabla(L_f h_i)^\top \mathbf{d}. \quad (18)$$

where $L_f h_i = \nabla h_i^\top f$, $L_f^2 h_i = \nabla(L_f h_i)^\top f$, and $L_g L_f h_i = \nabla(L_f h_i)^\top g$. All gradients are taken with respect to the joint state $\mathbf{x}$. For the squared-error barrier, the nominal derivatives can be computed from the coordination error:

$$\begin{aligned} L_f h_i &= -2\mathbf{e}_i^\top L_f \mathbf{e}_i, \\ L_f^2 h_i &= -2\|L_f \mathbf{e}_i\|^2 - 2\mathbf{e}_i^\top L_f^2 \mathbf{e}_i, \\ L_g L_f h_i &= -2\mathbf{e}_i^\top L_g L_f \mathbf{e}_i. \end{aligned} \quad (19)$$

The componentwise bounds $|\mathbf{d} - \hat{\mathbf{d}}| \le \boldsymbol{\varepsilon}$ imply

$$\nabla(L_f h_i)^\top (\mathbf{d} - \hat{\mathbf{d}}) \ge -\left|\nabla(L_f h_i)\right|^\top \boldsymbol{\varepsilon}. \quad (20)$$

Applying this bound to (18) and substituting into $\ddot{h}_i + (k_1 + k_2)\dot{h}_i + k_1 k_2 h_i \ge 0$ yields (13).

Assume that the payload-pose map is twice continuously differentiable and the estimation-error bounds hold along the trajectory. A feasible control enforcing (13) continuously gives $\dot{\psi}_i \ge -k_2 \psi_i$. For initial conditions $h_i(0) \ge 0$ and $\psi_i(0) \ge 0$, the resulting bounds are

$$\psi_i(t) \ge e^{-k_2 t}\psi_i(0) \ge 0, \qquad h_i(t) \ge e^{-k_1 t} h_i(0) \ge 0. \quad (21)$$

Thus, the joint state remains in $\mathcal{C}$ when these conditions hold for all carriers.

B. Active Coordination and Input Bounds

Differentiating $\dot{\mathbf{e}} = L_f \mathbf{e}$ and replacing the disturbance with its DOB estimate gives

$$\hat{\ddot{\mathbf{e}}} = L_f^2 \mathbf{e} + L_g L_f \mathbf{e} \, \mathbf{u} + \frac{\partial (L_f \mathbf{e})}{\partial \mathbf{x}} \hat{\mathbf{d}}. \quad (22)$$

The active coordination term in (14) penalizes deviations from the target error acceleration $-k_d \dot{\mathbf{e}} - k_p \mathbf{e}$. Since the prediction is affine in $\mathbf{u}$, this penalty is quadratic.

With the input and DOB estimate held constant over Δt, the set $\mathcal{U}$ imposes the following bounds:

$$\begin{aligned} &|u_{i,1}| \le a_i^{\max}, \qquad |u_{i,2}| \le \sin \beta_i^{\max}, \\ &\left|v_i + \Delta t (u_{i,1} + \hat{d}_{v,i})\right| \le v_i^{\lim} - \Delta t \varepsilon_v, \end{aligned} \quad (23)$$

where $a_i^{\max}$, $\delta_i^{\max}$, and $v_i^{\lim}$ are the acceleration, steering-angle, and speed limits, respectively, and $\beta_i^{\max} = \arctan\left(\frac{l_{r,i}}{l_{f,i}+l_{r,i}} \tan \delta_i^{\max}\right)$. The speed margin $\Delta t \varepsilon_v$ accounts for the integrated estimation error over one control interval.

The objective weights normalize the control corrections and error accelerations using the vehicle limits:

$$\mathbf{W}_i = \operatorname{diag}\left(\frac{1}{(a_i^{\max})^2}, \frac{1}{\sin^2 \beta_i^{\max}}\right), \mathbf{Q}_i = \frac{\mathbf{I}_2}{(a_i^{\max})^2}, \quad (24)$$

where $\mathbf{I}_2$ is the 2×2 identity matrix. The global matrices $\mathbf{W}$ and $\mathbf{Q}$ are assembled block-diagonally, with $\|\mathbf{y}\|_{\mathbf{W}}^2 = \mathbf{y}^\top \mathbf{W} \mathbf{y}$.

The optimized inputs are converted to acceleration and steering commands through

$$\begin{aligned} a_i^\star &= u_{i,1}^\star, \qquad \beta_i^\star = \arcsin u_{i,2}^\star, \\ \delta_i^\star &= \arctan\left(\frac{l_{f,i} + l_{r,i}}{l_{r,i}} \tan \beta_i^\star\right). \end{aligned} \quad (25)$$

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